

FINAL REVIEW

一、MULTIPLE CHOICE *15 (30 MARKS)

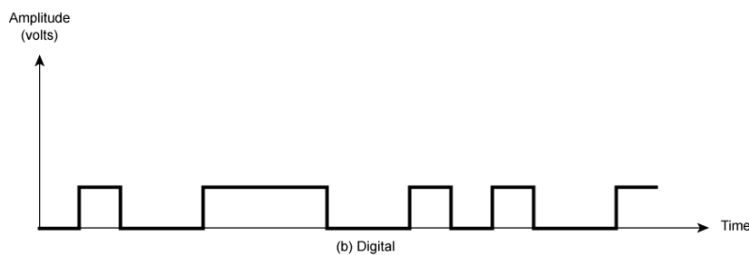
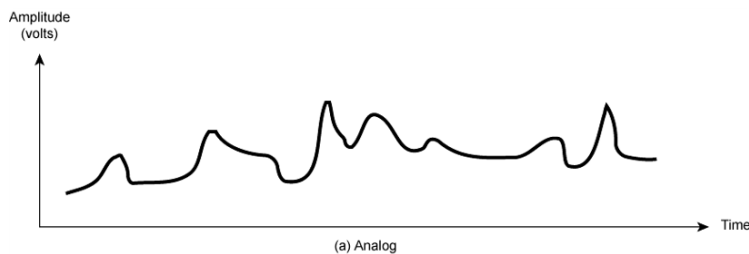
1. Analog / Digital data

Difference

Electromagnetic signal

Analog signal intensity varies smoothly with no breaks. -- Continuous

Digital signal intensity maintains a constant level and then suddenly changes to another level. – Discrete



2. Analog / Digital Transmission

Analog **data** -- codec 解码器 -> Digital bit stream

Digital **data** -- modem 调制解调器 -> Analog **signal**

Achieve longer distance,

- **Amplifiers** that boost the energy in the **signal** are used for **analog transmission (M)**
- **Repeaters** are used for **digital** transmission. A repeater receives the digital signal, **recovers** the pattern of 1s and 0s, and retransmit a new signal.

Analog transmission: Transmit analog signals – Analog data / digital data

- With amplifiers cascaded 串联 to achieve long distances, the signal becomes more and more distorted 变形

模拟传输适用于传输模拟信号。级联放大器增强信号的同时会增强噪声，对模拟数据的损伤较数字数据小。

- Analog data (voice)
- Digital data (binary data)

Digital transmission: Transmit digital signals.

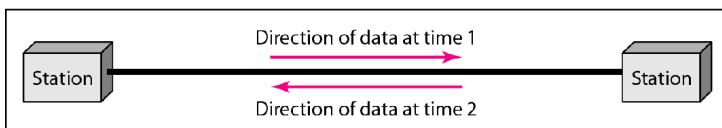
- A digital signal can be transmitted only a limited distance because of attenuation, noise, and other impairments 损伤 endanger the integrity of the data.

中继器可复原失真衰减的数字信号，中继器也可用于携带数字数据的模拟信号

- The repeater recovers the digital data from the analog signal and generates a new, clean analog signal. Thus noise is not cumulative.

3. Half Duplex (HD)

Both stations transmit, but only one at a time, The entire capacity of the channel can be used for each direction.



Used in cases -- no need for communication in both directions at the same time.

4. The network core --> TWO switching

You need to understand the difference between the two switching

Mesh of interconnected routers

Packet-switching: Hosts break application-layer messages into packets

- Forward packets from one router to the next, across links on path from source to destination
- Each packet transmitted at full link capacity – 每个数据包以链路全容量传输
 - So, Sending a packet of L bits over a link with transmission rate R bits/sec, then the time to transmit the packet is L/R seconds.

- Resource Not reserved. – 在分组交换网络中，这些资源不保留；会话的消息按需使用资源，因此，可能必须等待（即排队）才能访问通信链路。

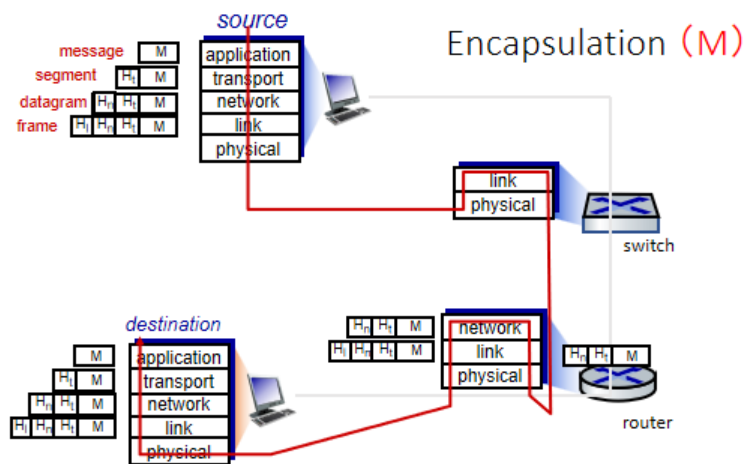
Circuit-switching:

- End-end resources allocated to, reserved for "call" between source and destination
- No-sharing – If not used by call -> idle
- Traditional telephone networks
- Resources reserved

Packet	Circuit
Internet	traditional telephone
No call setup/ termination phase	Call setup/termination phase
share	circuit allocated to each user (no exclusive 独有的)

- Call setup phase (to establish the circuit)
- Call termination phase (to release the circuit)

5. Encapsulation –封装



6. HTTP connections

non-persistent HTTP

- at most one object sent over TCP connection
 - connection then closed
- downloading multiple objects required multiple connections

persistent HTTP (default) (M)

- multiple objects can be sent **over single TCP** connection between client, server

HTTP uses persistent connections in its default mode, HTTP clients and servers can be configured to use non-persistent connections instead.

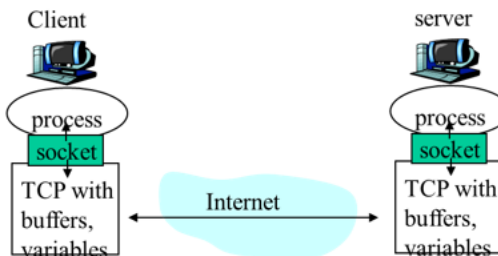
7. HTTP overview

*uses **TCP**: (M)*

- client initiates TCP connection (creates socket) to server, port 80
- server accepts TCP connection from client
- HTTP messages (application-layer protocol messages) exchanged between browser (HTTP client) and Web server (HTTP server)
- TCP connection closed

HTTP is “stateless”

- server maintains no information about past client requests



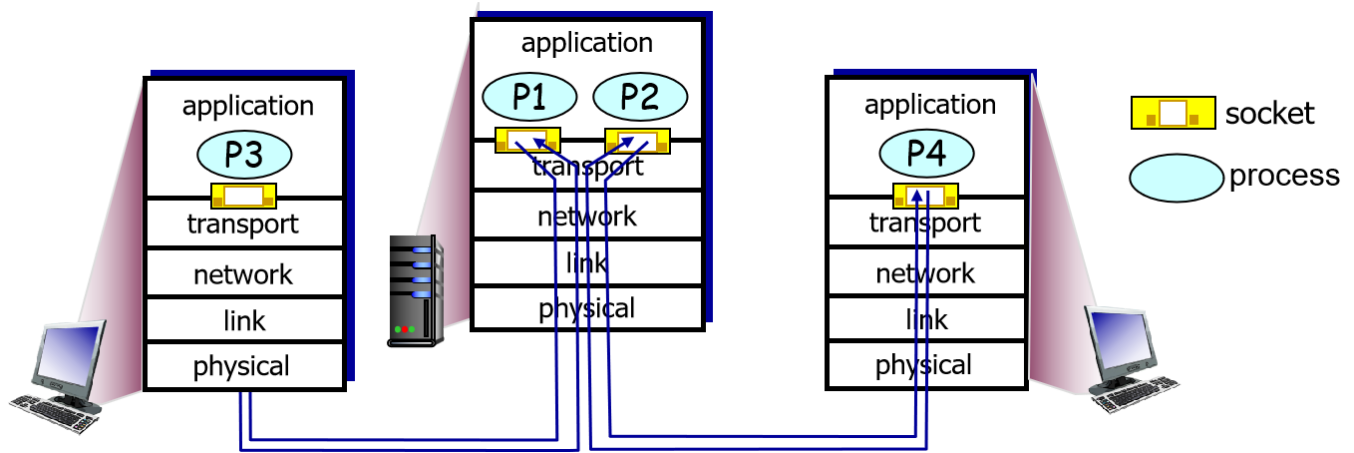
TCP provides a **reliable** data transfer service to HTTP. => INTACT! 完好无损

HTTP uses TCP as its underlying transport protocol.

The HTTP client first initiates a TCP connection with the server. Once the connection is established, 浏览器和服务端进程通过它们的套接字接口访问TCP。

服务器将请求的文件发送给客户机，而不存储有关客户机的任何状态信息。因此HTTP被称为**无状态协议**。

8. Multiplexing/ Demultiplexing



Socket:

- Act as door, network <=> Process
- A host can have one or more sockets.
- A host has a **unique IP address (32-bit for IPV4)**
- Each socket has a **port number (16-bit)** at a host, ranging from 0 to 65535.
 - The port numbers ranging from 0 to 1023 are called well-known port numbers and are restricted, which are reserved for use by well-known application protocols such as HTTP (port number 80) and FTP (port number 21). (M)

9. How to decide the sequence No.

❖ Example:

- The data file consisting of 500,000 bytes, MSS is 1000bytes, the initial sequence number is 0.
- TCP constructs **500 segments**; the **sequence number** set in the **first, second, third segments** is **0, 1000, 2000**, respectively.

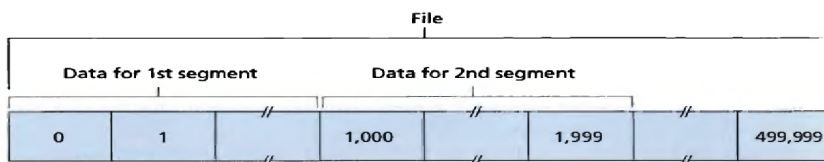
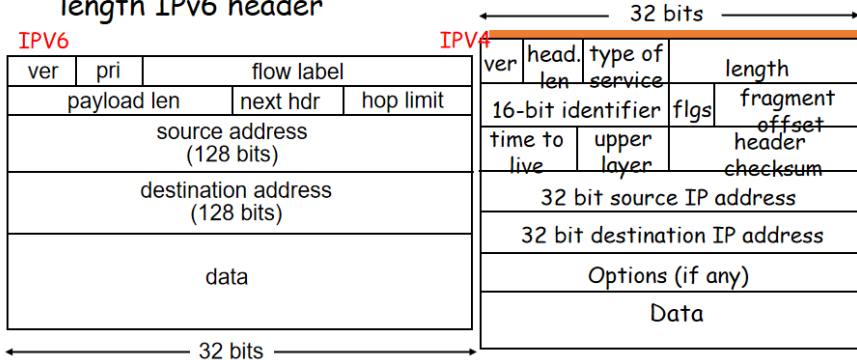


Figure 3.30 ♦ Dividing file data into TCP segments

- TCP views data as **ordered stream of bytes**.
- Sequence number is the number of the first data in the segment in unit of bytes. **Sequence numbers are over bytes, not segments.**

10. IPv6 Header

- ❑ **Large address space:** 32 bits to 128 bits
- ❑ **No fragmentation** in IPv6
- ❑ **No checksum field** in order to reduce processing time at each hop
- ❑ **No options field** in IPv6, which leads to 40-byte fixed length IPv6 header



11. Routing algorithm classification

Q: global or decentralized information?

global:

- all routers have complete topology, link cost info
- “link state” algorithms

decentralized:

- router knows physically-connected neighbors, link costs to neighbors
- iterative process of computation, exchange of info with neighbors
- “distance vector” algorithms

Q: static or dynamic?

static:

- ❖ routes **change slowly** over time

dynamic:

- ❖ routes **change more quickly**

- periodic update
- in response to link cost changes

- **Global routing algorithms** – All routers have complete information about the node connectivity (i.e., network topology), and link cost information.
 - Global routing algorithms use these **complete, global information** to compute the least-cost path between a source router to a destination router.“
 - “link state” algorithm <=> a global routing algorithm
- **Decentralized routing algorithms** – At the beginning, each router knows the cost of its directly connected links. Using the **iterative calculation** and **information exchange with the neighboring nodes**
 - A router gradually obtains the least-cost path to a destination router.

- "Distance vector" algorithm $\Leftrightarrow$ decentralized routing algorithm

12. IP address of subnet

- Dotted-decimal notation: a.b.c.d/x, where the notation /x is known as **subnet mask**, it indicate that x **leftmost bits of the 32-bit IP address** is the **subnet part**, and **remaining (32-x) bits** is the **host part (M)**

二、BIG QUESTION *9

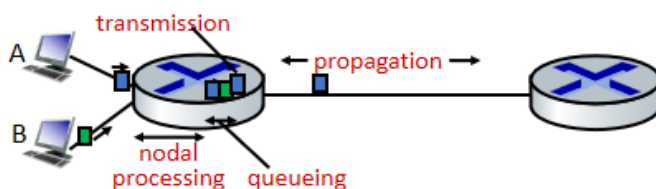
1. Transmission impairments 损伤

- Analog $\rightarrow$ impairment(Random modification) $\rightarrow$ Degradation 恶化 of signal quality
- Digital $\rightarrow$ Bit error

Most significant impairments are ?

- Attenuation 衰减
- Distortion 变形, 失真
- Noise

2. Packet delay : Four sources



$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

- 传播 Propagation delay : 信号在物理媒介 (如光纤) 上传播所需的时间, 与物理距离和传播速度相关
- 传输 Transmission delay : 将数据包推送到链路上所需的时间, 与数据包大小与链路带宽相关
- 排队延迟 queueing delay: 数据包在路由器或交换机中**排队等待**传输所需时间
- 处理延迟 Processing delay: The time a router / switch takes to examine the packet header and decide how to forward it.

- be able to caculate:

d_{trans} : transmission delay: (Q)

- L : packet length (bits)
- R : link transmission rate (bps)

▪ $d_{trans} = L/R$

d_{prop} : propagation delay: (Q)

- d : length of physical link
- s : propagation speed ($\sim 2 \times 10^8$ m/sec)

▪ $d_{prop} = d/s$

d_{trans} and d_{prop}
very different

- be able to explain:

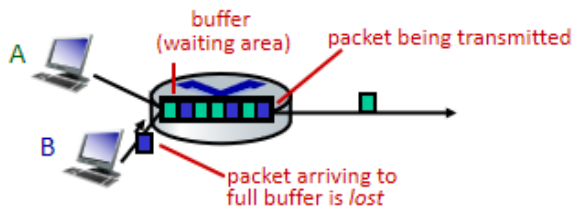
d_{proc} : nodal processing

- check bit errors
- determine output link
- typically < msec

d_{queue} : queueing delay (Q)

- time waiting at output link for transmission
- depends on congestion level of router

3. Packet loss (What is?)



Queue (aka buffer) preceding link in buffer has finite capacity

Packet arriving to full queue **dropped** (aka lost)

Lost packet may be retransmitted by previous node, by source end system, or not at all.

两者需解释含义（对比）

Queue Delay: 当路由器/交换机的缓冲区中挤压了大量的数据包时，新到的数据包需要排队等待被处理或传输，这会导致延迟。≠ 数据包丢失

- 延迟 => 排队等待 --> **缓冲区未滿，数据包不会丢失**
- 丢包 => **缓冲区滿，数据包丢失，数据无法传递**

Packet Loss: 当缓冲区已滿且无法容纳更多数据包时，新到的数据包就会被丢弃；通常发生在网络拥塞严重的情况下，队列超过了缓冲区的容量限制

4. Packet switching versus circuit switching (Q)

Packet switching allows more users to use network

Circuit switching – Wasteful – The dedicated circuits 专用电路 are idle during silent periods.

- **Why internet use packet switching instead of Circuit switching?**

- a. **Efficiency:**

- Packet Switching – Data is broken into small packets, which are sent independently and routed through the network. => Network resources are dynamically shared among multiple users.
 - Circuit Switching – A dedicated path is reserved for the entire duration of the communication, even when there is no data being transmitted. => Wasted resources during idle periods.

- b. **Scalability :**

- Packet switching – Allows multiple users to share the same infrastructure. It adapts well to growing numbers of devices and users
 - Circuit switching – Struggles to scale, as it requires dedicated pathways for each communication session.

- c. **Flexibility:**

- Packet switching – Enables data to be sent along the most efficient routes. If one path is congested or fails, packets can be rerouted dynamically, ensuring delivery.
 - Circuit switching – Lacks this flexibility—if the dedicated path fails, the connection is lost entirely.

- d. **Cost Effectiveness:**

- Since packet switching **optimizes the use of network resources** and **doesn't require dedicated connections**, it's more cost-effective to implement and maintain.

- **Is packet switching a 'slam dunk winner'? (Q)**

Great for 'Bursty' data — sometimes user has data to send, but at other times not

- Packet switching: resource sharing
- Circuit switching: dedicated and exclusive

5. Transport service requirements: common apps

TCP/ UDP connection – **full-duplex**.

TCP service:

- **connection-oriented**: setup required between client and server processes
- **reliable transport**: to deliver all data sent without error and in the proper order.
- **flow control**: sender won't overwhelm receiver
- **congestion control**: throttle sender when network overloaded
- **does not provide**: timing, minimum throughput guarantee, security

UDP service:

- **connectionless**, so there is no handshaking before the two processes start to communicate.
- **unreliable data transfer**
- **does not provide**: reliability, flow control, congestion control, timing, throughput guarantee, security, or connection setup,

UDP or TCP ? => Offers a different set of services to different app

TCP Connection-oriented service. => **Handshaking**

When the application finishes sending messages, it must tear down the connection.

TCP Reliable data transfer service. => **Without error and in the proper order.**

当发送方和接收方之间的网络拥塞时，TCP拥塞控制机制会对发送方进程（客户端或服务器）进行节流。

UDP is a no-frills, lightweight transport protocol, providing minimal services. UDP is connectionless => **No** handshaking before the two processes start to communicate.

UDP provides an unreliable data transfer service => **No** guarantee that the message will ever reach the receiving process. & Out of order.

And we also know that TCP can be easily enhanced at the application layer with SSL to provide security services. TCP可以很容易地在应用层通过SSL进行增强，以提供安全服务。

COMMON APPS – WHAT TRANSPORT PROTOCOL?

application	data loss	throughput	time sensitive
file transfer	no loss	elastic	no
e-mail	no loss	elastic	no
Web documents	no loss	elastic	no
real-time audio/video	loss-tolerant	audio: 5kbps-1Mbps video: 10kbps-5Mbps	yes, 10s of msec
stored audio/video	loss-tolerant	same as above	yes, few secs
interactive games	loss-tolerant	few kbps up	yes, 10s of msec
text messaging	no loss	elastic	yes and no

TCP: 应用于需要高可靠性和数据完整性的场景（如文件下载、网页浏览、电子邮件等）。

UDP: 应用于实时性高或对数据丢失容忍度较高的场景（如流媒体、视频会议、在线游戏）。

特性	TCP	UDP
拥塞控制	有，动态调整吞吐量	无，固定发送速率
可靠性	高，牺牲部分吞吐量换取可靠性	无，仅传输数据不管丢失
实时性	较低，适合非实时应用	高，适合实时传输

TCP强调可靠性=>吞吐量弹性变化， TCP 会根据网络状态动态调整吞吐量

Internet transport protocols services

application	application layer protocol	transport protocol
file transfer/download	FTP [RFC 959]	TCP
e-mail	SMTP [RFC 5321]	TCP
Web documents	HTTP 1.1 [RFC 7320]	TCP
Internet telephony	SIP [RFC 3261], RTP [RFC 3550], or proprietary	TCP or UDP
interactive games	WOW, FPS (proprietary)	UDP or TCP
Real time video streaming (zoom,voIP)		UDP
stored streaming(Netflix, you tube)		TCP

分析的重点：可靠传输（loss>吞吐量弹性变化） & Time sensitive

1. 可靠性:

TCP: 提供端到端的可靠性，通过确认机制（ACK）、重传和序列号确保数据完整性和正确顺序。=> 高可靠性

UDP: 不提供可靠性保障，数据包可能丢失或乱序，适合对丢包或乱序不敏感的应用（如视频流、在线游戏）。

2. 实时性和延迟:

TCP: 因为有拥塞控制、确认机制和握手过程，TCP 的延迟较高，不太适合低延迟、实时性强的应用。

UDP: 延迟低，无需握手或确认，适用于实时性要求高的应用，如语音通信、直播。

3. 流量特性:

TCP: 面向连接，适用于需要持续双向通信的数据流（如 HTTP、HTTPS 流量）。

UDP: 面向无连接，更适合短时突发流量或广播/多播的应用场景（如 DNS 查询、IPTV）。

4. 网络拥塞:

TCP: 内置拥塞控制和流量控制机制，适合在拥挤的网络中传输数据。

UDP: 没有拥塞控制，数据包无论网络状况如何都会发送，因此在拥堵情况下可能丢包严重，但在带宽充足的情况下效率高。

5. 资源消耗:

TCP: 对系统资源要求较高（如内存、CPU），因其需要维护连接状态、重传队列等。

UDP: 资源占用低，适合在硬件资源有限或高并发环境中使用。

6. UDP checksum/ Internet checksum

What is the object of checksum?

How to use?

Goal: Detect "errors" (e.g., flipped bits) in transmitted segment

DETECT!!!! ≠ 重传

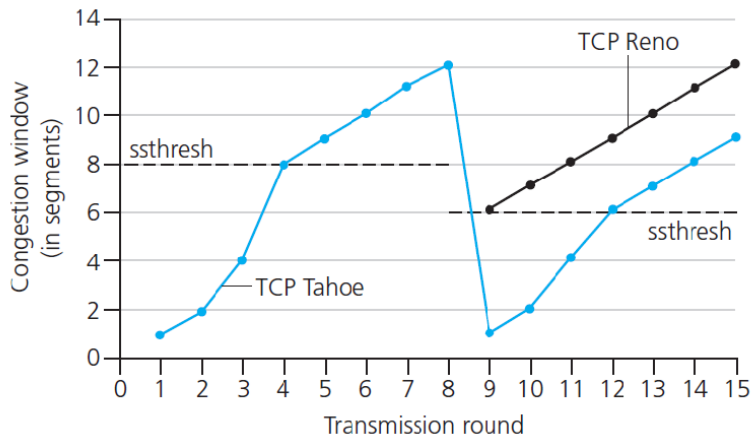
- TCP – 通过确认机制（ACK）和重传确保数据可靠性。如果检测到错误 => 重传
- UDP – 使用校验和检测数据是否损坏，但不提供重传或纠正机制，损坏的数据包会被丢弃。

example: add two 16-bit integers

	1	1	1	0	0	1	1	0	0	1	1	0	0	1	1	0
	1	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1
wraparound	1	1	0	1	1	1	0	1	1	1	0	1	1	1	0	1
sum	1	0	1	1	1	0	1	1	1	0	1	1	1	1	0	0
checksum	0	1	0	0	0	1	0	0	0	1	0	0	0	0	1	1

7. TCP: switching from slow start to CA

Calculate how many RTT it takes from start to any other phase?



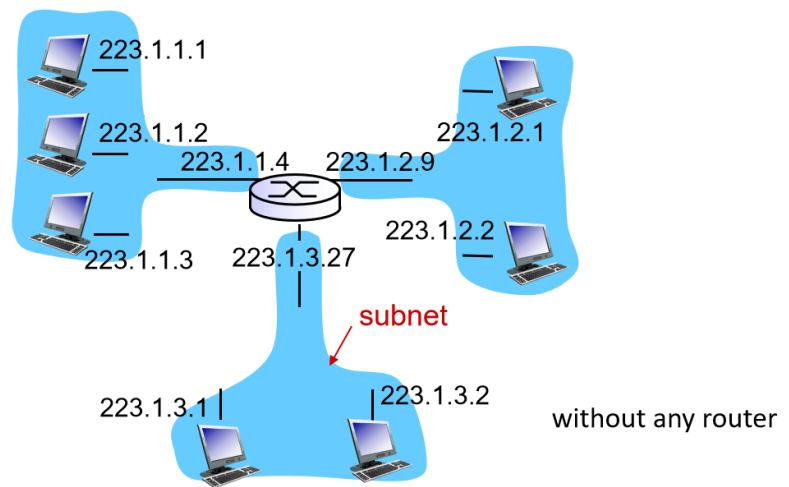
TCP Tahoe 和 TCP Reno 的区别在于丢包后行为：

- TCP Tahoe：完全回到慢启动阶段，窗口减小到最小值。
- TCP Reno：在轻微拥塞时直接进入拥塞避免阶段，不回到慢启动。

8. Subnets –There is a big Q related to subnet.

Subnets (Q)

- The structure of IP address:
 - subnet part - high order bits
 - host part - low order bits
- *what's a subnet ?*
 - A subnet is a network where interfaces can physically reach each other without passing through an intervening router



network consisting of 3 subnets

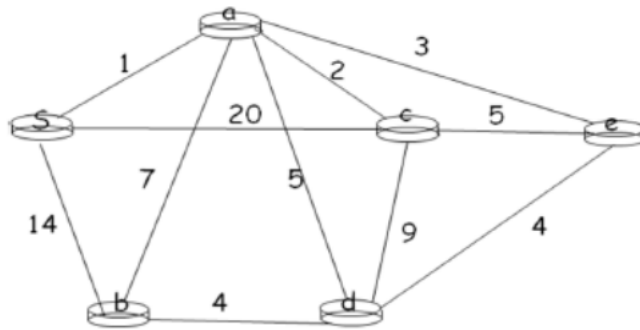
□ IP address of subnet

- Dotted-decimal notation: a.b.c.d/x, where the notation /x is known as **subnet mask**, it indicate that **x leftmost bits of the 32-bit IP address** is the **subnet part**, and **remaining (32-x) bits** is the **host part (M)**
- Interfaces in the same subnet have the same subnet part of IP address
- Example: the above network includes three subnets with subnet mask /24.
- The subnet mask is used by the TCP/IP protocol to determine whether a host is on the local subnet or on a remote network.

1

9. Chpter 5 – exercise 1

1. The figure below shows the structure of a network.
 - a) Please use Dijkstra's algorithm to find the least-cost paths from the node S to other nodes a, b, c, d and e, respectively.
 - b) Note that you need to give the process of each iteration.



▼ ANS

The first iteration:

we start with the source node $N'=\{s\}$ --表明N中元素

To get the cost of other nodes:

the adjacent nodes of s: $D(A)=1; D(B)=14; D(C)=20;$ --写出COST (相邻/不相邻)

all other nodes: $D(D)=\infty; D(e)=\infty;$

Comparing all nodes without the set N' , we find the node with the **minimum cost D** is a, then add node a into the set N' . **We have $N'=\{s,a\}$** , and we find the least cost path from s to a is s-a with the cost 1. --找最短COST, 把那个结点添加到N中

The second iteration:

To get the cost of nodes **without the set N'** :

Update the distance for the adjacent nodes of a: --更新除N中的其他结点的COST

$D(B)=\text{Min}[D(B), D(A)+C(A,B)]=\text{Min}[14, 1+7]=8$ --找出Min(源路径, 使用刚刚得出的路径+剩余路径)

同理, $D(C)=3; D(D)=6; D(e)=4$

Comparing all nodes without the set N' , we find the node with the **minimum cost D** is c, then **add node c into the set N'** . We have $N'=\{s,a,c\}$, and we find the least cost path from s to c is s-a-c with the cost 3.

The third iteration:

To get the cost of nodes without the set N' :

Update the distance for the adjacent nodes of c: $D(D)=6; D(e)=4;$ --相邻的找Min

The cost of any other nodes keep unchanged: $D(B)=8;$ --不相邻的不用管

Comparing all nodes without the set N' , we find the node with the **minimum cost D** is e, then **add node e into the set N'** . We have $N'=\{s,a,c,e\}$, and we find the least cost path from s to e is s-a-e with the cost 4.

The fourth iteration:

To get the cost of nodes without the set N' :

Update the distance for the adjacent nodes of e: $D(D) = 6$;

The cost of any other nodes keep unchanged: $D(B) = 8$;

Comparing all nodes without the set N' , we find the node with the **minimum cost D** is d, then **add node d into the set N'** . We have $N' = \{s, a, c, e, d\}$, and we find the least cost path from s to d is s-a-d with the cost 6.

The fifth iteration:

To get the cost of nodes without the set N' :

update the distance for the adjacent nodes of d: $D(B) = 8$;

Since we only have one node without the set N' ,

we have the least cost path from s to b is s-a-b with the cost 8.

Then we add node b into the set N' . We have $N' = \{s, a, c, e, d, b\}$.

Show the forwarding table:

a) The forwarding table of node S is:

a	(s,a)
b	(s,a)
c	(s,a)
d	(s,a)
e	(s,a)

展示第一跳!

三、未明确指出考察选择/大题的知识点

1. How does a host get IP address?

a. Hard-coded by system admin in a file

b. DHCP (Dynamic Host Configuration Protocol) – Dynamically get address from as server.

2. DHCP goal: Allow host to dynamically obtain its IP address from network server when it joins network
3. DHCP function:
 - a. Dynamically obtain its IP address
 - b. Address of first-hop router for client
 - c. Name and IP address of DNS sever
 - d. network mask (indicating network versus host portion of address (e.g. 255.255.255.0))

▼ 解释

a. 动态获取设备的 IP 地址

- 每个设备在加入网络时需要一个唯一的 IP 地址。DHCP 自动为设备分配 IP 地址，避免手动配置，减少冲突风险。
- 当设备断开网络时，该 IP 地址会被回收并重新分配给其他设备。

b. 为客户端提供第一跳路由器的地址

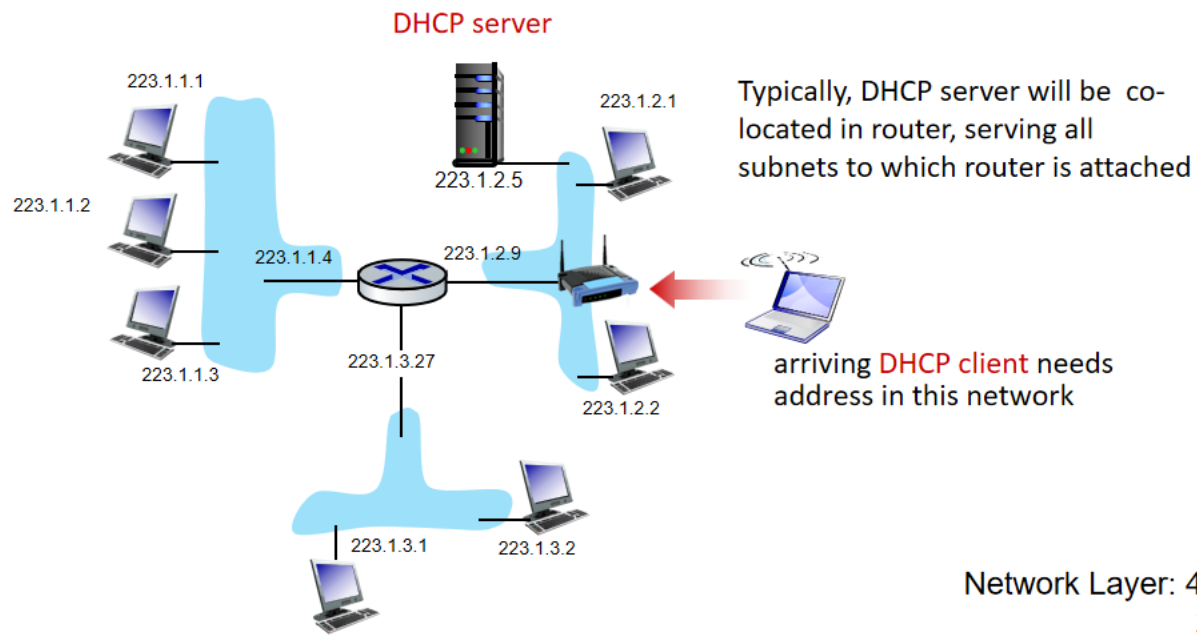
- 第一跳路由器（也称网关）是设备与其他网络（如互联网）通信的出入口。
- DHCP 会将这个路由器的 IP 地址发送给客户端，确保设备可以正确发送数据包到外网。

c. 提供 DNS 服务器的名称和 IP 地址

- DNS（域名系统）用于将人类可读的域名（例如 www.example.com）解析为设备可以识别的 IP 地址。
- DHCP 会自动告知设备使用哪个 DNS 服务器进行域名解析，无需手动配置。

d. 提供网络掩码

- 网络掩码（例如 255.255.255.0）用于定义 IP 地址中哪一部分是网络部分，哪一部分是主机部分。
- 它帮助设备区分同一网络内的设备与外部设备，从而正确路由数据。
- DHCP 服务器预先设置好网络掩码（例如 **255.255.255.0**），与每个网络地址范围绑定。
- 当设备向 DHCP 服务器发送请求时，服务器会将网络掩码作为一部分配置参数发送给设备。



当设备加入网络时：

1. **DHCP Discover:** 客户端通过广播向 DHCP 服务器发出请求，希望获得网络配置信息。
2. **DHCP Offer:** DHCP 服务器响应，提供 IP 地址、网关、DNS 服务器和网络掩码等信息。
3. **DHCP Request:** 客户端选择其中一个 DHCP Offer，发送确认请求。
4. **DHCP Acknowledgement:** DHCP 服务器确认分配，并将所有网络配置（包括网络掩码）发送给客户端。